

Question	Answer	Marks	Guidance
9(a)	difference between mass of nucleus and mass of constituent nucleons	M1	Allow 'protons and neutrons' for 'nucleons'. Mass must be mentioned once.
	when separated to infinity	A1	Ignore 'mass equivalence of binding energy'.
9(b)(i)	alpha decay identified as producing Ra-224 (plus He-4)	B1	
	β^+ decay identified as producing Ac-228 and β^- decay as producing Pa-228	B1	Must be the correct way around.
	alpha decay products calculated to have total mass defect of $1.877217u$	B1	Any appearance of the value $1.877217u$ is sufficient to award the mark.
	reaction only happens if mass defect of products is greater than mass defect of parent nucleus (so decay is alpha and is not beta)	B1	
9(b)(ii)	$E = (\Delta)mc^2$	C1	
	$= (1.877217 - 1.871290) \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$	C1	1.877217 may be left as 1.846840 + 0.030377. 1.877217 – 1.871290 may be substituted as 0.005927 Allow subtraction either way round Treat use of 1.67×10^{-27} as a TE in this C1 mark (gives an answer of 5.57 MeV).
	$(= 8.85 \times 10^{-13} \text{ J})$ $= 5.53 \text{ MeV}$	A1	Correct 3 SF answer in J implies both C1 marks. Must be given to 3 significant figures. Answer must be positive for A1.